Inspecting failures and anomalies

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete research-case evidence record.

Separate valid slow behavior, demonstrated measurement errors and untested explanations.

Keep one habit: Recall → W1–W3/W5 → independent case → project evidence → feedback and delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Running the primary experiment. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Power and resource tradeoffs. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Privacy and data governance. Give an example and a mistake that the idea helps you avoid.

Running the primary experiment

My remembered idea, example and correction

Power and resource tradeoffs

My remembered idea, example and correction

Privacy and data governance

My remembered idea, example and correction

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Failure inspection generates hypotheses to test, not automatic reasons to erase unusual cases. Main valid durations 1, 2, 2, 15 seconds have sum twenty, mean five and median two. The slow valid case remains in the declared primary analysis. Deleting it without a rule would give mean 5/3 seconds, an illustration of distortion rather than an approved result.

  Separate picture durations 2, 2, 2, 10 have mean four and median two. Fourth-run logs show a source timeout and the plan includes timeouts. Removing ten because it looks bad violates that plan. A logged timeout is not a complete root-cause explanation. Inspect raw outputs, input validity, timing boundaries and procedure traces.

  Correct demonstrated measurement errors transparently with originals and provenance; retain valid unusual behavior. New post-result investigations are exploratory until tested. No actual timing run, timeout or causal experiment is performed here.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

- Failure inspection generates hypotheses to test, not automatic reasons to erase unusual cases. Main valid durations 1, 2, 2, 15 seconds have sum twenty, mean five and median two. The slow valid case remains in the declared primary analysis. Deleting it without a rule would give mean 5/3 seconds, an illustration of distortion rather than an approved result.

  Separate picture durations 2, 2, 2, 10 have mean four and median two. Fourth-run logs show a source timeout and the plan includes timeouts. Removing ten because it looks bad violates that plan. A logged timeout is not a complete root-cause explanation. Inspect raw outputs, input validity, timing boundaries and procedure traces.

  Correct demonstrated measurement errors transparently with originals and provenance; retain valid unusual behavior. New post-result investigations are exploratory until tested. No actual timing run, timeout or causal experiment is performed here.

- Durations 1, 2, 2, 15 seconds are all valid traces. Team deletes 15 as an outlier without a rule. Show the intermediate steps and give the conclusion with its scope.

Main calculation	Complete arithmetic	Interpretation
Retained durations	1, 2, 2, 15 seconds	All valid under supplied contract
Sum / count	20 seconds / 4	No undeclared exclusion
Mean	20/4 = 5 seconds	Sensitive to the long valid duration
Median	Middle values 2 and 2; average 2 seconds	Different descriptive aspect
Wrong-deletion illustration	(1+2+2)/3 = 5/3 seconds	Shows distortion; not a replacement primary result
Cause of fifteen-second duration	Not supplied	Propose an investigation without inventing its outcome

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Failure inspection generates hypotheses to test, not automatic reasons to erase unusual cases. Main valid durations 1, 2, 2, 15 seconds have sum twenty, mean five and median two. The slow valid case remains in the declared primary analysis. Deleting it without a rule would give mean 5/3 seconds, an illustration of distortion rather than an approved result.

  Separate picture durations 2, 2, 2, 10 have mean four and median two. Fourth-run logs show a source timeout and the plan includes timeouts. Removing ten because it looks bad violates that plan. A logged timeout is not a complete root-cause explanation. Inspect raw outputs, input validity, timing boundaries and procedure traces.

  Correct demonstrated measurement errors transparently with originals and provenance; retain valid unusual behavior. New post-result investigations are exploratory until tested. No actual timing run, timeout or causal experiment is performed here.

- A report omits this qualification: Post-hoc exclusion based on an unwanted outcome can distort the conclusion. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Failure inspection generates hypotheses to test, not automatic reasons to erase unusual cases. Main valid durations 1, 2, 2, 15 seconds have sum twenty, mean five and median two. The slow valid case remains in the declared primary analysis. Deleting it without a rule would give mean 5/3 seconds, an illustration of distortion rather than an approved result.

  Separate picture durations 2, 2, 2, 10 have mean four and median two. Fourth-run logs show a source timeout and the plan includes timeouts. Removing ten because it looks bad violates that plan. A logged timeout is not a complete root-cause explanation. Inspect raw outputs, input validity, timing boundaries and procedure traces.

  Correct demonstrated measurement errors transparently with originals and provenance; retain valid unusual behavior. New post-result investigations are exploratory until tested. No actual timing run, timeout or causal experiment is performed here.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Values 2, 3, 4, 20 are valid. Compute mean and median and propose an honest investigation.
- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- Add a labelled inspecting failures and anomalies **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of inspecting failures and anomalies.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Inspecting failures and anomalies. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Inspecting failures and anomalies. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Values 2, 3, 4, 20 are valid. Compute mean and median and propose an honest investigation.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Inspect an **anomaly** without erasing it. A timing study records four runs and notices one unusually long duration.

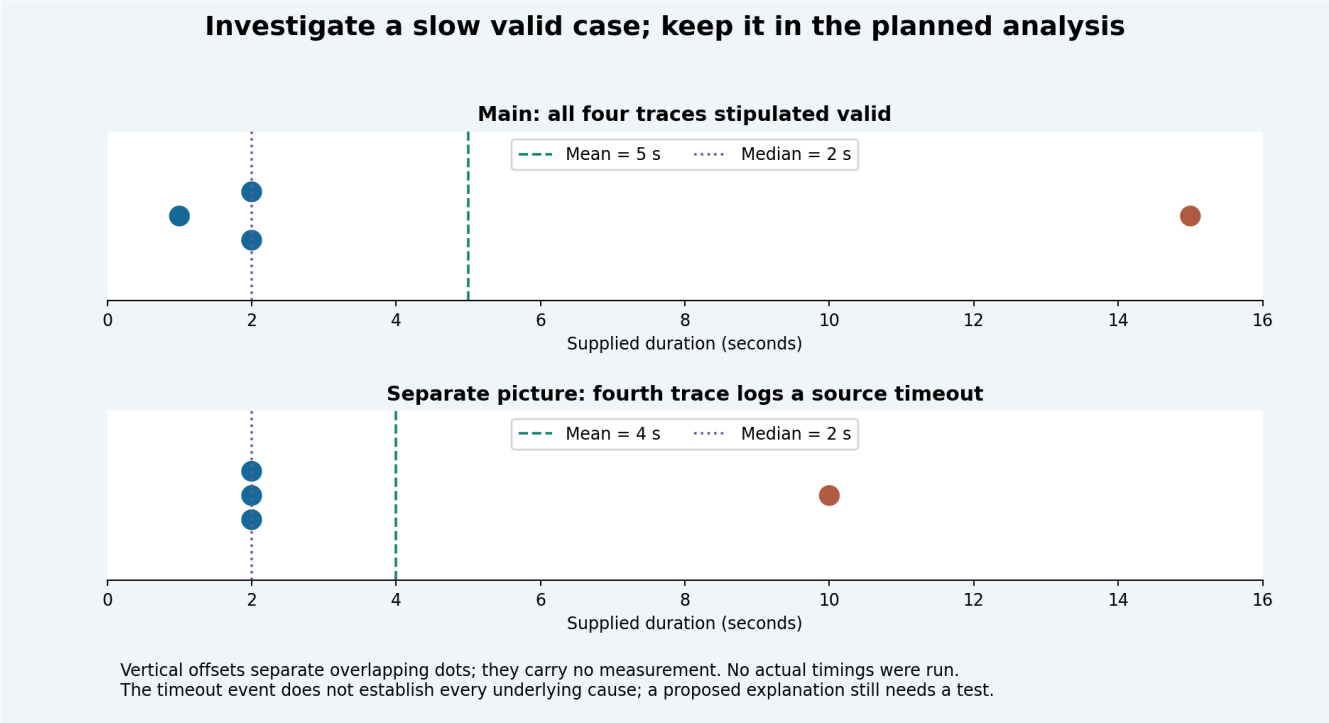
- Durations are 2,2,2,10 seconds. Logs show a source timeout on the fourth run. The plan includes timeouts in reporting.

- Calculate mean and median for 2,2,2,10. What does the timeout log establish, and why is deleting that inconvenient record incompatible with the stated plan?

- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

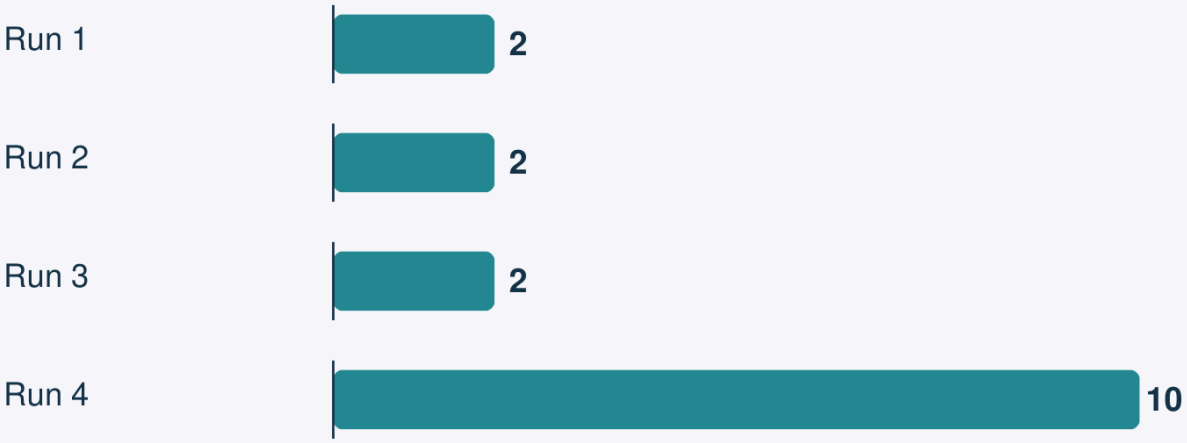
  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.



AI ACADEMY / AI-7/26

All four recorded durations

seconds; run 4 includes timeout



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/26

Inspecting failures and anomalies

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Keep the raw record

The fourth duration remains 10 seconds.



2. Compute summaries

Mean is 16/4=4 seconds; median is 2 seconds.



3. Investigate the event

Report the timeout evidence and distinguish it from an untested explanation.

CHANGE ONE THING

Delete the 10-second record only because it looks bad.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Inspect an **anomaly** without erasing it. A timing study records four runs and notices one unusually long duration.
- Which labelled step matches this detail? Report the timeout **evidence** and distinguish it from an untested explanation.
- 1. Investigate the event
- 2. Keep the raw record
- 3. Compute summaries
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Investigate the event Report the timeout **evidence** and distinguish it from an untested explanation. An **anomaly** invites investigation, not automatic deletion.

A slow failure may be part of real service behaviour; the predefined rule determines inclusion.

Changed-case question and optional hint

- Changed-case practice: Delete the 10-second record only because it looks bad. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. The mean becomes 2 seconds but the selective report violates this plan.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The mean becomes 2 seconds but the selective report violates this plan. Observed inconvenient outcomes are not an approved exclusion reason.

Repair and retain the record

Review Running the primary experiment (ai-7/25). Goal: Explain the **mechanism** and terminology of running the primary experiment. Short practice: Compute A's and B's six-case counts and identify the cases that favour each.

How must a B revision tuned on case four be labelled when evaluated again on the same set?. Return to this week and explain what you changed.

Review Power and resource tradeoffs (ai-7/23). Goal: Explain the **mechanism** and terminology of power and resource tradeoffs. Short practice: Calculate both simulated detection fractions and resource costs.

What additional detection is observed for the extra forty units, and does that establish power for an actual classroom study?. Return to this week and explain what you changed.

Review Privacy and data governance (ai-7/20). Goal: Explain the **mechanism** and terminology of privacy and data governance. Short practice: Point to the analysis table and the restricted identity key.

Why are the codes pseudonymous rather than guaranteed anonymous, and what changes if the key is published?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

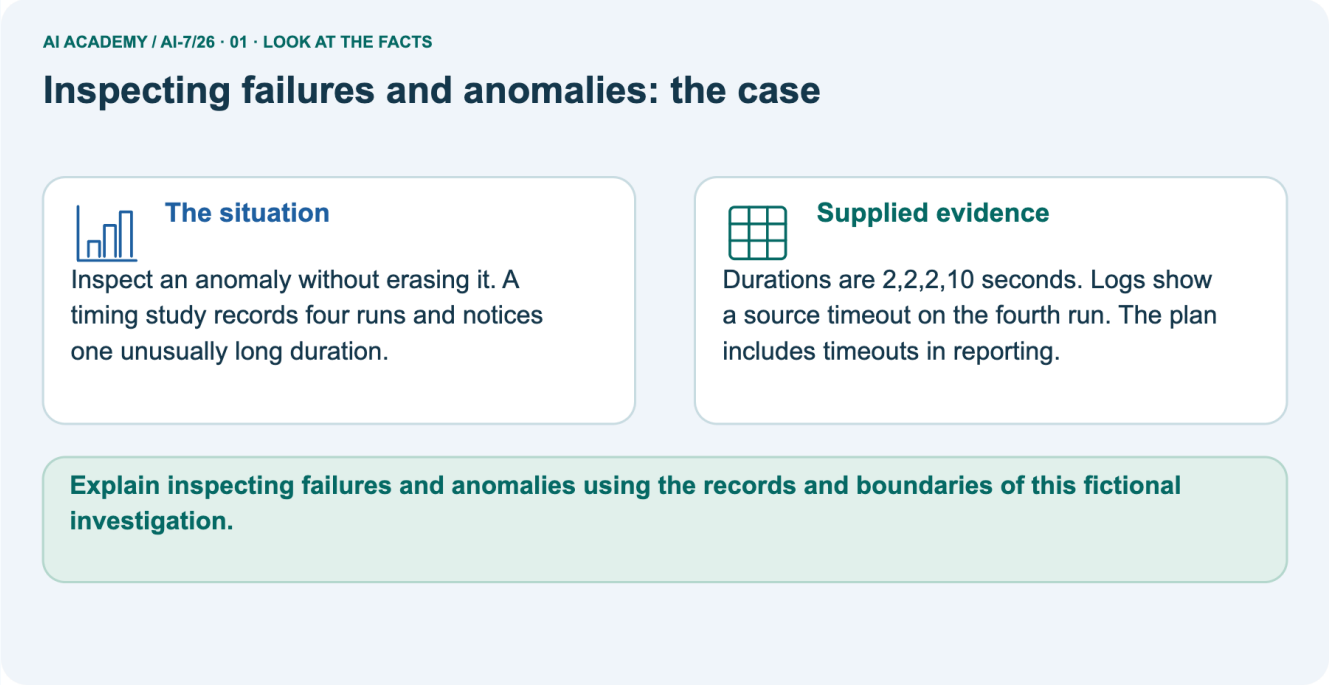
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Inspecting failures and anomalies: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

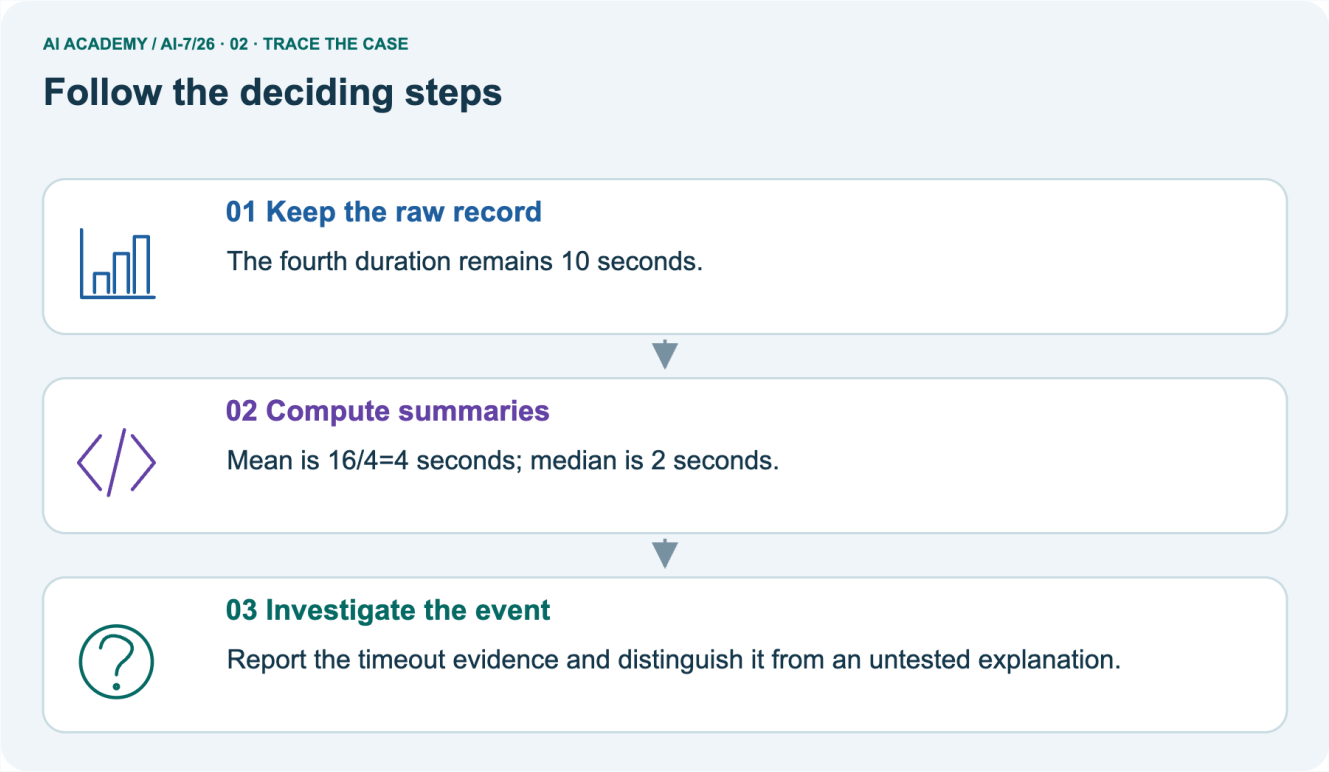
Inspect an **anomaly** without erasing it. A timing study records four runs and notices one unusually long duration. Supplied facts: Durations are 2,2,2,10 seconds. Logs show a source timeout on the fourth run.

The plan includes timeouts in reporting.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



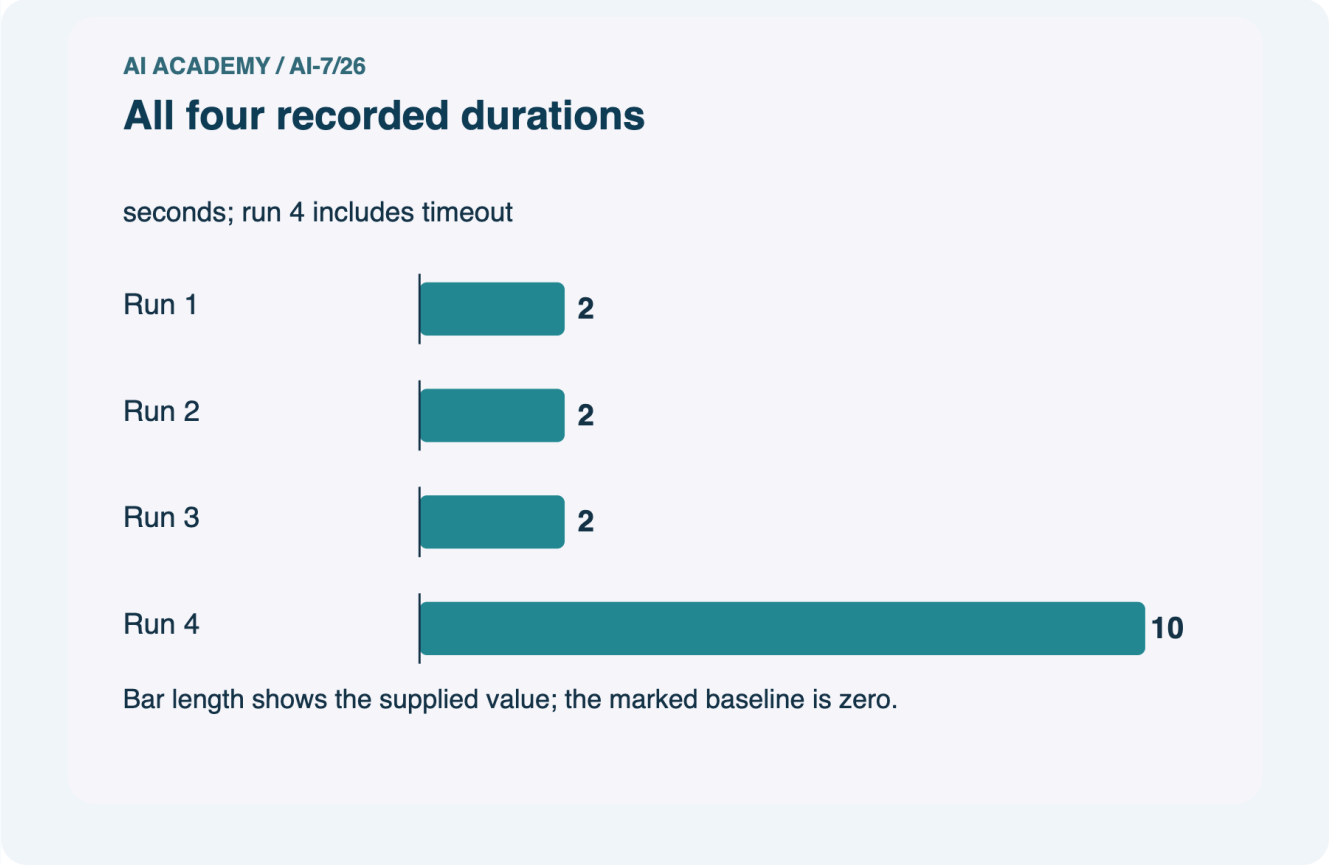
Follow the labelled picture with the matching explanation. Open the vector diagram

Keep the raw record: The fourth duration remains 10 seconds. → Compute summaries: Mean is 16/4=4 seconds; median is 2 seconds. → Investigate the event: Report the timeout **evidence** and distinguish it from an untested explanation.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. All four recorded durations



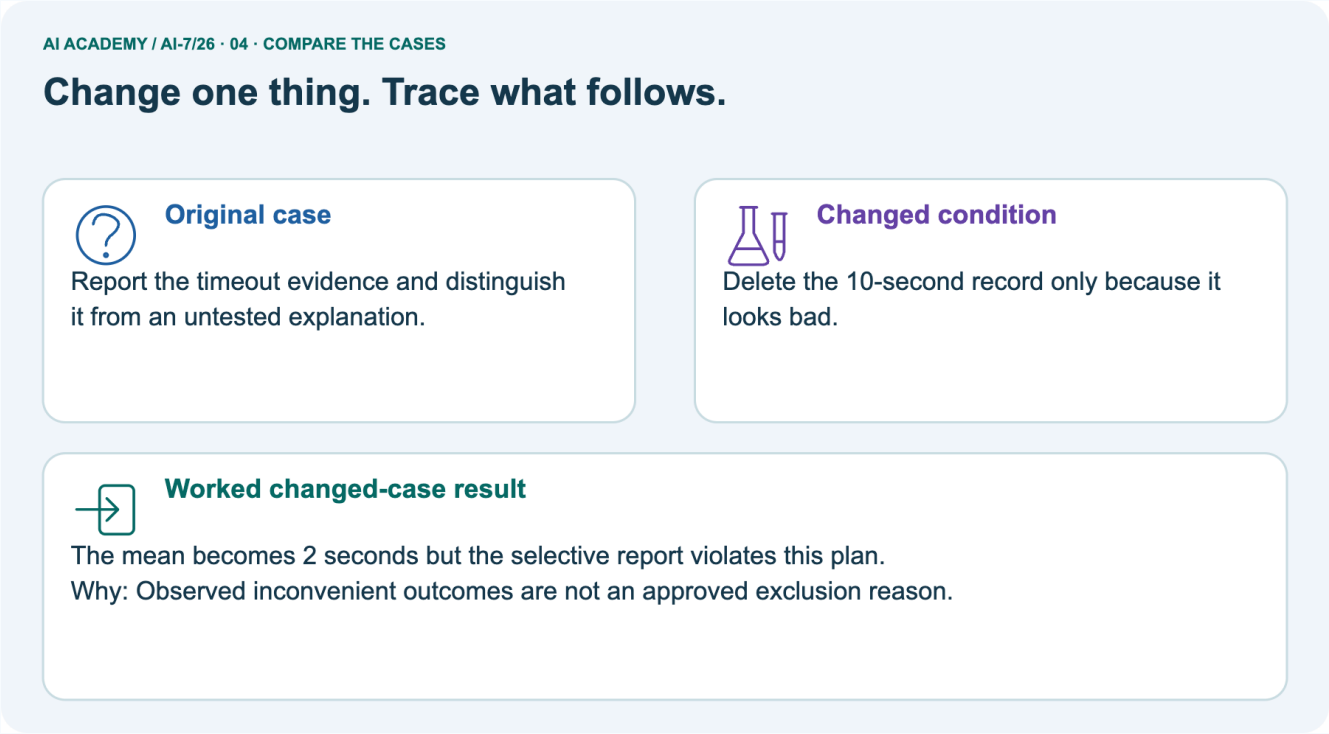
Follow the labelled picture with the matching explanation. Open the vector diagram

All four recorded durations. Units: seconds; run 4 includes timeout. Run 1: 2. Run 2: 2. Run 3: 2. Run 4: 10.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

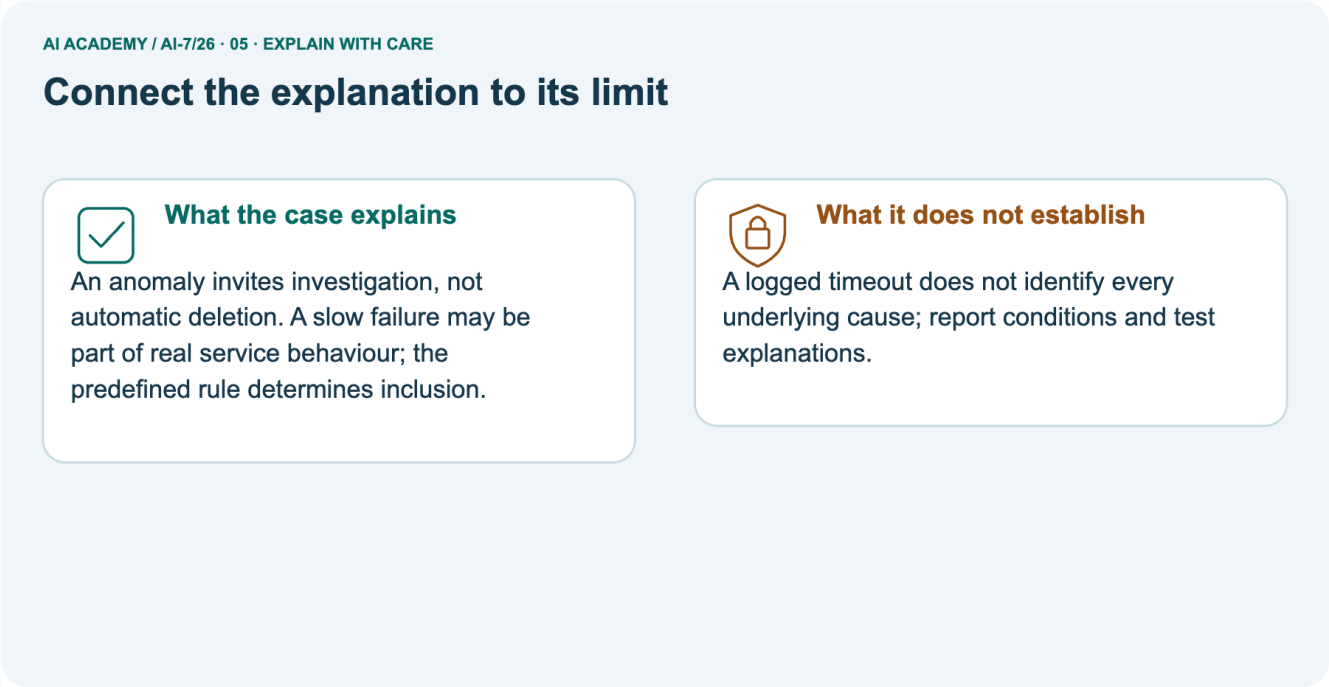
Original case: Report the timeout **evidence** and distinguish it from an untested explanation.
Changed condition: Delete the 10-second record only because it looks bad.

Expected result: The mean becomes 2 seconds but the selective report violates this plan.
Reason: Observed inconvenient outcomes are not an approved exclusion reason.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

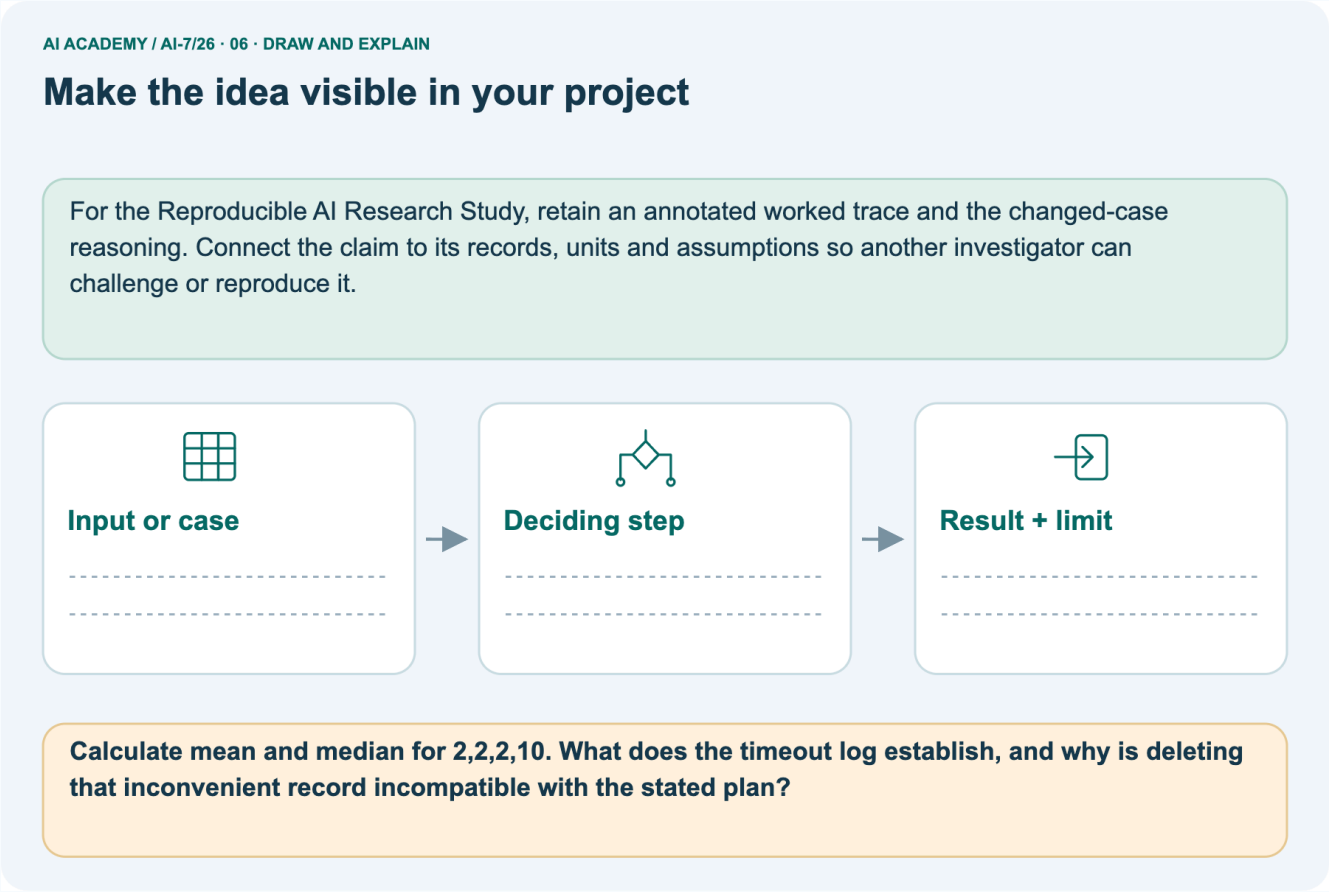
An **anomaly** invites investigation, not automatic deletion. A slow failure may be part of real service behaviour; the predefined rule determines inclusion.

Limit: A logged timeout does not identify every underlying cause; report conditions and test explanations.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate mean and median for 2,2,2,10. What does the timeout log establish, and why is deleting that inconvenient record incompatible with the stated plan?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Inspecting failures and anomalies: the case

02. Follow the deciding steps

03. All four recorded durations

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.